

A Short Guide to Proposition C.3

The multimode least-noise moment

Abstract

Proposition C.3 says that the quantum-limited amplifier maximizes every product of decreasing photon-number tests among displacement-covariant amplifiers of the same gain. It allows a trace-nonincreasing map and an arbitrary multimode map that may couple the modes. The proof first reduces to the one-mode stochastic-order theorem of Guță and Matsumoto, then turns the other modes into weighted environments and iterates the one-mode bound. This note explains the normalization, phase twirl, stochastic order, and tensorization argument.

1 Informal meaning

Fix an intensity gain $\gamma > 1$. A displacement-covariant amplifier must send an input displacement z to the amplified displacement $\sqrt{\gamma}z$, but it may add different amounts of noise. The quantum-limited amplifier, whose idler is in the vacuum, adds the least possible noise. After phase twirling, a displacement-covariant amplifier can be represented by the same two-mode-squeezing interaction with different phase-invariant idler states. Photons already present in the idler create additional stimulated output photons; choosing the vacuum removes this avoidable contribution, leaving only the spontaneous noise required by amplification. The least-noise theorem strengthens this mean-noise intuition by showing that the entire vacuum-idler photon-number law is smallest in stochastic order.

For a phase-invariant output state ρ with photon-number probabilities $P_k = \langle k|\rho|k\rangle$, the expectation of a number test is

$$\mathrm{Tr}[\rho w(\hat{N})] = \sum_{k \geq 0} P_k w(k).$$

If w is decreasing, an outcome with fewer photons receives a larger score; for example, $w(k) = \mathbf{1}_{\{k \leq K\}}$ measures the probability of a low-energy outcome, while $w(k) = e^{-sk}$ smoothly discounts high photon numbers. Moving probability mass toward higher photon numbers can only decrease such an expectation. Hence the least-noise output, whose photon-number law is stochastically smallest, has the largest expectation of $w(\hat{N})$ among the admissible amplifiers. In several modes, Proposition C.3 proves the corresponding statement for product tests

$$W(\hat{\mathbf{N}}) = \prod_{\ell=1}^s w_{\ell}(\hat{N}_{\ell}).$$

The competitor Γ need not factor over the modes; the proof is designed to handle correlations created by a joint channel.

2 Thermal inputs and the reference amplifier

On one bosonic mode, the thermal state with parameter $q \in (0, 1)$ is

$$\tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle\langle k|. \quad (1)$$

Its photon-number distribution is geometric. At gain γ , define

$$q^{(\gamma)} = 1 - \frac{1 - q}{\gamma}. \quad (2)$$

The quantum-limited amplifier maps

$$\tau_q \longmapsto \tau_{q^{(\gamma)}}.$$

Since

$$q^{(\gamma)} - q = \frac{(\gamma - 1)(1 - q)}{\gamma} > 0,$$

the output geometric law is broader than the input law, as amplification requires.

For s modes, choose $q_1, \dots, q_s \in (0, 1)$ and write

$$\tau = \bigotimes_{\ell=1}^s \tau_{q_\ell}, \quad \tau^{(\gamma)} = \bigotimes_{\ell=1}^s \tau_{q_\ell^{(\gamma)}}. \quad (3)$$

3 Decreasing tests and stochastic order

A probability law P on $\mathbb{Z}_{\geq 0}$ stochastically dominates Q if

$$\sum_k f(k) P_k \geq \sum_k f(k) Q_k$$

for every bounded increasing function f . Equivalently,

$$\sum_k w(k) P_k \leq \sum_k w(k) Q_k \quad (4)$$

for every bounded nonnegative decreasing function w . Equivalently, writing $Q \preceq_{\text{st}} P$,

$$\sum_{k=0}^m Q_k \geq \sum_{k=0}^m P_k \quad (m \geq 0).$$

The direction is easy to remember: the stochastically larger law places more mass at high photon numbers, so it scores less under a test that rewards low photon numbers. Proposition C.3 uses this scalar order for the photon-number distributions of phase-invariant states; it does not assert an operator inequality between density matrices.

4 Statement of Proposition C.3

Proposition C.3 (rephrased). *Let Γ be a completely positive trace-nonincreasing map on the s -mode trace class satisfying*

$$\Gamma(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^* \quad (z \in \mathbb{C}^s).$$

If each $w_\ell : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}_+$ is bounded and decreasing, then

$$\text{Tr} \left[\Gamma(\tau) \prod_{\ell=1}^s w_\ell(\hat{N}_\ell) \right] \leq \text{Tr}[\Gamma(\tau)] \text{Tr} \left[\tau^{(\gamma)} \prod_{\ell=1}^s w_\ell(\hat{N}_\ell) \right]. \quad (5)$$

To explain the factor $\text{Tr} \Gamma(\tau)$ in (5), abbreviate

$$W = W(\hat{N}) = \prod_{\ell=1}^s w_{\ell}(\hat{N}_{\ell}) \quad \text{and} \quad c = \text{Tr} \Gamma(\tau).$$

The positive operator $\Gamma(\tau)$ has trace c . If $c = 1$, it is an ordinary normalized output state. If $0 < c < 1$, it is a subnormalized state. Here the probabilistic operation is the map Γ itself: it is completely positive and trace-nonincreasing, and c is its probability of success on the input τ . Conditional on success, the normalized output is

$$\rho_{\text{out}} = \frac{\Gamma(\tau)}{c}.$$

Substituting $\Gamma(\tau) = c\rho_{\text{out}}$ into the two sides of (5) gives

$$c \text{Tr}[\rho_{\text{out}} W] \leq c \text{Tr}[\tau^{(\gamma)} W].$$

After cancelling c , this is precisely the usual least-noise comparison between two normalized states. The word “homogeneous” refers only to this common factor: replacing Γ by $a\Gamma$ multiplies both sides of (5) by the same number a . If $c = 0$, positivity gives $\Gamma(\tau) = 0$, and both sides are zero.

There is a second reason for allowing $c < 1$ in Proposition C.3. The maps constructed later in Appendix C are weak-* limits of channels. Such a limit can lose trace. For example,

$$\rho_j = |j\rangle\langle j| \quad \text{has} \quad \text{Tr} \rho_j = 1, \quad \text{Tr}(\rho_j K) \longrightarrow 0 \quad \text{for every compact } K.$$

Thus ρ_j converges weak-* to the zero operator, whose trace is zero: the states move to higher and higher photon numbers, beyond what any fixed compact observable detects. The phrase “trace lost at infinity” refers to exactly this phenomenon. More generally, if channels Γ_j converge weak-* to Γ , then on the input τ one may have

$$\text{Tr} \Gamma_j(\tau) = 1 \quad \text{for every } j, \quad \Gamma_j(\tau) \xrightarrow{w^*} \Gamma(\tau), \quad c := \text{Tr} \Gamma(\tau) < 1.$$

For $c > 0$, write $\Gamma(\tau) = c\rho_{\text{out}}$, where $\text{Tr} \rho_{\text{out}} = 1$. Equation (5) then reads

$$\underbrace{\text{Tr}[\Gamma(\tau) W]}_{c \text{Tr}[\rho_{\text{out}} W]} \leq \underbrace{\text{Tr} \Gamma(\tau)}_c \text{Tr}[\tau^{(\gamma)} W].$$

Thus the two appearances of c are exactly the two appearances of $\Gamma(\tau)$ in (5): once inside its weighted expectation on the left and once through its trace on the right.

5 Proof: the one-mode case

Step 1: phase twirling changes neither side

Let $R_{\theta} = e^{i\theta \hat{N}}$. Replace Γ by

$$\Gamma^{\text{ph}}(X) = \frac{1}{2\pi} \int_0^{2\pi} R_{-\theta} \Gamma(R_{\theta} X R_{\theta}^*) R_{-\theta}^* d\theta.$$

The thermal input and $w(\hat{N})$ commute with every R_{θ} , so both traces in (5) are unchanged. The twirl preserves complete positivity, trace nonincrease, and displacement covariance, and it makes the map phase covariant. Independent phase twirls are used in the multimode case.

Phase covariance is what lets the proof speak about the output photon-number law and makes the idler state diagonal in its number basis.

Step 2: covariance makes the success probability constant

The positive output-trace functional has the form

$$\mathrm{Tr} \Gamma(X) = \mathrm{Tr}(AX), \quad 0 \leq A \leq I.$$

The displacement covariance relation is

$$\Gamma(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^*.$$

Taking the trace and using invariance of the trace under unitary conjugation gives

$$\mathrm{Tr} \Gamma(D(z)XD(z)^*) = \mathrm{Tr} \Gamma(X).$$

Using $\mathrm{Tr} \Gamma(Y) = \mathrm{Tr}(AY)$ on both sides and cyclicity of the trace, this becomes

$$\mathrm{Tr}[AD(z)XD(z)^*] = \mathrm{Tr}(AX), \quad \text{hence} \quad \mathrm{Tr}[(D(z)^*AD(z) - A)X] = 0$$

for every trace-class operator X . Therefore

$$D(z)^*AD(z) = A \quad (z \in \mathbb{C}).$$

The Weyl representation is irreducible, so Schur's lemma gives

$$A = cI, \quad \mathrm{Tr} \Gamma(X) = c \mathrm{Tr} X, \quad 0 \leq c \leq 1. \quad (6)$$

If $c = 0$, positivity forces $\Gamma = 0$ and the desired inequality is trivial. If $c > 0$, the normalized map

$$\Lambda = c^{-1}\Gamma$$

is a trace-preserving channel satisfying phase covariance and displacement covariance with gain $\sqrt{\gamma}$.

Step 3: apply the least-noise stochastic theorem

The arbitrary-gain Weyl-operator characterization used in Sec. 3 and the Appendix of Guță–Matsumoto writes

$$\Lambda^*(D(\xi)) = \mathrm{Tr}[\sigma D(-\sqrt{\gamma-1}\bar{\xi})]D(\sqrt{\gamma}\xi) \quad (7)$$

for an idler state σ . Phase covariance makes σ diagonal in the number basis. In the two-mode-squeezer realization, the output annihilation operator has the form

$$a_{\mathrm{out}} = \sqrt{\gamma} a_{\mathrm{in}} + \sqrt{\gamma-1} b_{\mathrm{idler}}^*.$$

Consequently, for phase-invariant input and idler states,

$$\mathbb{E}N_{\mathrm{out}} = \gamma \mathbb{E}N_{\mathrm{in}} + (\gamma-1)(\mathbb{E}N_{\mathrm{idler}} + 1),$$

so the vacuum idler, for which $N_{\mathrm{idler}} = 0$, clearly minimizes the mean output photon number. The result needed here is stronger than this mean comparison: Guță and Matsumoto's number-state calculation shows that every nonzero idler occupation shifts the entire output law upward in first-order stochastic order. Since a diagonal idler state is a mixture of number states, its output law also stochastically dominates the vacuum-idler law $\tau_{q^{(\gamma)}}$.

For the thermal input, set

$$P_{\Lambda}(k) = \langle k | \Lambda(\tau_q) | k \rangle, \quad P_{\gamma}(k) = (1 - q^{(\gamma)})(q^{(\gamma)})^k.$$

The arbitrary-gain stochastic-order theorem states precisely that the vacuum-idler law P_γ is stochastically smallest:

$$P_\gamma \preceq_{\text{st}} P_\Lambda, \quad \sum_{k=0}^m P_\gamma(k) \geq \sum_{k=0}^m P_\Lambda(k) \quad (m \geq 0). \quad (8)$$

Thus “smallest photon number” refers to the whole distribution, not to a deterministic photon number or a pointwise comparison of probabilities. Summation by parts converts the cumulative inequalities in (8) into the decreasing-test inequality: if $F_P(m) = \sum_{k=0}^m P(k)$ and $w_\infty = \lim_m w(m)$, then

$$\sum_{k \geq 0} P(k)w(k) = w_\infty + \sum_{m \geq 0} (w(m) - w(m+1))F_P(m).$$

All coefficients in the last sum are nonnegative, so $F_{P_\gamma} \geq F_{P_\Lambda}$ gives

$$\text{Tr}[\Lambda(\tau_q)w(\hat{N})] \leq \text{Tr}[\tau_{q(\gamma)}w(\hat{N})].$$

Multiplying by $c = \text{Tr} \Gamma(\tau_q)$ proves (5) for one mode.

The stochastic minimality in this step is the specialized external input. The phase twirl, constant-trace reduction, normalization, and conversion to the conic inequality are internal parts of the proof.

6 Proof: the multimode reduction

One cannot simply multiply the one-mode inequalities because Γ may couple all modes. Instead, the proof packages the other modes into a weighted environment.

If one of the weights is identically zero, both sides of (5) vanish. Otherwise rescale each weight by its positive supremum; by homogeneity it is enough to assume

$$0 \leq w_\ell \leq 1.$$

Set

$$W_{>1} = \prod_{\ell=2}^s w_\ell(\hat{N}_\ell), \quad \tau_{>1} = \bigotimes_{\ell=2}^s \tau_{q_\ell},$$

and define a one-mode map

$$\Gamma_1(X) = \text{Tr}_{2,\dots,s} \left[(I \otimes W_{>1}^{1/2}) \Gamma(X \otimes \tau_{>1}) (I \otimes W_{>1}^{1/2}) \right]. \quad (9)$$

Every operation in (9) is completely positive. Since $0 \leq W_{>1} \leq I$ and Γ is trace nonincreasing, Γ_1 is trace nonincreasing. It inherits displacement covariance in the first mode; after the independent phase twirl, it also inherits phase covariance. Apply the one-mode result to the map Γ_1 , the first-mode thermal input τ_{q_1} , and the first-mode test $w_1(\hat{N}_1)$. It gives

$$\text{Tr} \left[\Gamma_1(\tau_{q_1}) w_1(\hat{N}_1) \right] \leq \text{Tr} \Gamma_1(\tau_{q_1}) \text{Tr} \left[\tau_{q_1(\gamma)} w_1(\hat{N}_1) \right].$$

The left side is the original weighted expression, because

$$\begin{aligned} \text{Tr} \left[\Gamma_1(\tau_{q_1}) w_1(\hat{N}_1) \right] &= \text{Tr} \left[\Gamma(\tau_{q_1} \otimes \tau_{>1}) (w_1(\hat{N}_1) \otimes W_{>1}) \right] \\ &= \text{Tr}[\Gamma(\tau)W]. \end{aligned}$$

Here the first equality follows from the definition of the partial trace and cyclicity of the trace. Similarly, omitting the first-mode test gives

$$\mathrm{Tr} \Gamma_1(\tau_{q_1}) = \mathrm{Tr}[\Gamma(\tau)(I \otimes W_{>1})] = \mathrm{Tr}[\Gamma(\tau)W_{>1}].$$

Substituting these two identities into the one-mode bound yields

$$\mathrm{Tr}[\Gamma(\tau)W] \leq \mathrm{Tr}[\tau_{q_1}^{(\gamma)} w_1(\hat{N}_1)] \mathrm{Tr}[\Gamma(\tau)W_{>1}]. \quad (10)$$

Equation (10) removes the first weight and replaces it by the corresponding quantum-limited expectation. Repeat the same construction for modes $2, \dots, s$. After the last mode, the remaining term is $\mathrm{Tr} \Gamma(\tau)$, while the extracted scalar factors multiply to

$$\prod_{\ell=1}^s \mathrm{Tr}[\tau_{q_\ell}^{(\gamma)} w_\ell(\hat{N}_\ell)] = \mathrm{Tr}[\tau^{(\gamma)} W].$$

This is precisely (5).

7 The particular witness used later

For the orbital Gaussian model, let

$$\tau = \sum_{\mathbf{k}} t_{\mathbf{k}} |\mathbf{k}\rangle\langle\mathbf{k}|, \quad \tau^{(\gamma)} = \sum_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)} |\mathbf{k}\rangle\langle\mathbf{k}|.$$

Appendix C introduces

$$W_\gamma = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle\langle\mathbf{k}|. \quad (11)$$

Because both thermal laws are products,

$$\sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} = \prod_{\ell} \sqrt{\frac{1 - q_\ell}{1 - q_\ell^{(\gamma)}}} \left(\frac{q_\ell}{q_\ell^{(\gamma)}} \right)^{k_\ell/2}.$$

Each factor is positive, bounded, and decreasing. It also tends to zero as $k_\ell \rightarrow \infty$, so W_γ is compact and injective. Proposition C.3 applies to this witness, and direct calculation gives

$$\mathrm{Tr}(\tau^{(\gamma)} W_\gamma) = \mathrm{Tr}(\tau W_\gamma^{-1}) = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}} = B_{\mathrm{orb}}(\gamma, p). \quad (12)$$

Thus one witness simultaneously matches the weighted fidelity inequality, the compact-limit topology, and the least-noise moment bound.

8 Dependencies and role in the paper

The logical core is

$$\begin{aligned} & \text{Weyl irreducibility and one-mode least-noise stochastic order} \\ & + \text{mode-by-mode reduction} \implies \text{Proposition C.3.} \end{aligned}$$

The orbital specialization of Proposition C.5 applies it to the exactly covariant limit of Følner-averaged channels. The full classical–quantum specialization applies it to a classically weighted quantum limit after first proving that the limit is trace nonincreasing. In both cases, (12) converts the moment estimate into the factor $B_{\text{orb}}(\gamma, p)$.

Primary source for the one-mode stochastic theorem. M. Guță and K. Matsumoto, “Optimal cloning of mixed Gaussian states,” *Physical Review A* **74**, 032305 (2006), [arXiv:quant-ph/0605161](https://arxiv.org/abs/quant-ph/0605161).